

ALPER ALIMOĞLU, Ph.D.

Distributed Systems & Blockchain Researcher | Python Software Engineer

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PROFESSIONAL SUMMARY

Ph.D. in Computer Engineering with 12+ years across software development, blockchain systems, and distributed computing. Most of that in research; the recent roles are industry.

SKILLS

Languages: Python, Bash, Solidity **Frameworks & tools:** Flask, REST APIs, Docker
Platforms: Ethereum, GNU/Linux **Domains:** Blockchain, Distributed Computing, Smart Contracts

EXPERIENCE

Data Scientist @Odeabank Nov 2025-Present Istanbul, Turkey
Refactoring a Python backend project (~20k lines) that recommends optimal fund offers to customers; improved runtime performance.

Python Developer @Huawei Mar 2025-Aug 2025 Istanbul, Turkey
Fixed bugs in database Python projects. First-authored an academic paper.

Part-time Lecturer @Boğaziçi University Feb 2025-Jun 2025 Istanbul, Turkey
Taught *Functional Programming and Verification* to senior undergraduate students.

Middleware Administrator @Anadolu Hayat Emeklilik Aug 2024-Nov 2024 Istanbul, Turkey
Built a Microsoft Teams chatbot with a Python/Flask backend powered by Microsoft's LLM; deployed company-wide.

Cloud Engineer @ICRON A.Ş. Mar 2024-Apr 2024 Istanbul, Turkey
Managed the company's Azure cloud infrastructure; migrated databases across regions.

R&D Engineer @Boğaziçi University Feb 2017-Jul 2024 Istanbul, Turkey
Python backend development for own Ph.D. research; core contributor to eBlocBroker (github.com/ebloc/ebloc-broker).

Research Project Assistant @Research Foundation for SUNY Aug 2012-Jul 2015 NY, USA
Smart Load Balancer: Applied Echo State Network ML models for workload prediction in data centers.
b-ext4 File System: Hybrid SSD/HDD design to optimize performance and reduce energy consumption.

Internships @Cisco, Ericsson, ITU Robotics Lab 2009-2010 Istanbul, Turkey
Cisco University Prize-winning research intern. RoboCup 3D Simulation Team (ITU AI & Robotics Lab).

EDUCATION

Ph.D. Computer Engineering Boğaziçi University 2016-2024 GPA 3.64/4.0
Thesis: **EBlocBroker: Blockchain-based Computational Broker for e-Science** Advisor: Prof. Can Ozturan

M.Sc. Computer Science Binghamton University - SUNY 2012-2015 Valedictorian, GPA 3.77/4.0
Thesis: **Short-Term Workload Prediction with Reservoir Computing** Advisor: Prof. Kanad Ghose

B.Sc. Information Systems Engineering (Dual Diploma) Binghamton University-SUNY&ITU 2008-2012
Summa Cum Laude, GPA 3.89/4.0 (SUNY), Overall 3.62/4.0

PUBLICATIONS

Journal Articles

- Alimoglu, A., Özturan, C. "Autonomous Blockchain Organization for Sustainable Software and Execution History", *Blockchain: Research and Applications* (ESCI Q1), May 2026. Elsevier Link
- Alimoglu, A., Özturan, C. "An Autonomous Blockchain-Based Workflow Execution Broker for e-Science", *Cluster Computing*, Springer (SCI-E Q1), May 2024. DOI:10.1007/s10586-024-04534-z
- Alimoglu, A., Özturan, C. "An Autonomous Blockchain-Based Computational Broker for e-Science", *Concurrency and Computation: Practice and Experience*, Wiley (SCI-E Q1), Mar 2024. DOI:10.1002/cpe.8087

Conference Papers

- Alimoglu, A., Erdayandi, K., Mustafa, M. A., Cali, Ü. "EDGChain-E: A Decentralized Git-Based Framework for Versioning Encrypted Energy Data", IEEE International Conference on Distributed Ledger Technologies (ICDLT), Dec 2025. IEEE Xplore Link
- Alimoglu, A., Özturan, C. "Design of a Smart Contract Based Autonomous Organization for Sustainable Software", *IEEE 13th International Conference on e-Science*, Oct 2017. IEEE Xplore Link
- Shahriar Asta, Tuna Sonmez, Onuralp Ulusoy, Alper Alimoglu, Mustafa Ersen, Ozdemir Sozmen, Sanem Sariel-Talay. "BeeStanbul Team Description Paper", *RoboCup Symposium Workshop*, Istanbul, Jul 2011. Link

AWARDS & HONORS

- 🏆 **Valedictorian:** Graduate commencement speaker, Binghamton University, 2014, Speech: "Protect Our Dreams"
- ★ **Summa Cum Laude:** Top of Class, B.Sc. in Information Systems Engineering, 2012

- ★ **Dean's List:** Multiple honors at Binghamton University - SUNY & ITU
- ★ **Academic Achievement Award:** Binghamton University - SUNY, GPA 3.98/4.0, 2012
- 🏆 **Cisco University Prize:** Winner, 2010